

BAIO LEARNING SYSTEM

PU-II / NEET BIOLOGY COMPREHENSIVE REVISION NOTES

Chapter 4: Principles of Inheritance and Variation

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1. Introduction to Genetics & Historical Context

Genetics is the branch of biological science that explores **heredity** (transmission of traits from parents to offspring) and **variation** (the degree of phenotypic and genotypic difference among individuals of the same species and between offspring and parents).

Historical Understanding: Humans recognized as early as 8000–1000 B.C. that sexual reproduction was a key source of natural variation. Early farmers and breeders exploited natural variations present in wild ancestral species through selective breeding and artificial hybridization.

Classic Example: The development of high-yielding, domesticated Indian milch cattle such as **Sahiwal cows in Punjab** from ancestral wild cattle.

2. Essential Genetic Terminology & Key Concepts

- **Gene (Factor):** The fundamental physical and functional unit of heredity. It carries coded nucleotide sequences required to synthesize functional RNA/proteins determining specific traits. Mendel referred to genes as 'factors'.
- **Alleles:** Alternative/contrasting forms of the same gene residing at the same locus on homologous chromosomes (e.g., alleles T and t for plant height).
- **Locus (pl. Loci):** The fixed, specific physical coordinate or position of a gene on a chromosome.
- **Phenotype:** The observable physical, physiological, or biochemical manifestation of an organism (e.g., Tall vs. Dwarf, Violet vs. White flowers).
- **Genotype:** The complete genetic makeup, allelic combination, or chromosomal constitution of an individual (e.g., TT , Tt , tt).
- **Homozygous:** Presence of identical alleles at a given locus on homologous chromosomes (e.g., TT - homozygous dominant, tt - homozygous recessive). Pure-breeding line.
- **Heterozygous:** Presence of two dissimilar alleles at a given locus on homologous chromosomes (e.g., Tt). Organisms with this condition are called *hybrids*.
- **Dominant Allele:** An allele that phenotypically expresses itself in both homozygous (TT) and heterozygous (Tt) conditions, masking the alternative allele.
- **Recessive Allele:** An allele whose phenotypic expression is completely suppressed in the presence of a dominant allele; it expresses exclusively in the homozygous state (tt).
- **Monohybrid & Dihybrid Cross:** Crosses analyzing the inheritance pattern of one single character (e.g., stem height) or two pairs of contrasting characters (e.g., seed shape and colour) simultaneously.
- **Hemizygous:** The condition of having only a single copy of a gene/chromosome (e.g., X-linked genes in human males XY).

3. Mendel's Hybridization Experiments on Garden Pea (*Pisum sativum*)

Gregor Johann Mendel (1822–1884), honored as the *Father of Genetics*, conducted statistical hybridization experiments on garden peas (*Pisum sativum*) for seven years (**1856 – 1863**). His groundbreaking success was due to strict mathematical logic, large statistical sample sizes, selection of clear-cut contrasting characters, and meticulous artificial cross-pollination protocols (emasculatation and bagging).

Why Mendel Selected *Pisum sativum* (Garden Pea):

- Possesses distinctly discernible pairs of contrasting traits.
- Naturally self-pollinating with cleistogamous-like floral morphology, yet readily accessible for artificial cross-pollination.
- Short life cycle (annual plant) allowing analysis of multiple generations in a short span.
- High fecundity — produces a large number of seeds per cross, ensuring statistically robust sample sizes.
- Easy cultivation and maintenance in limited experimental garden plots.

Mendel's 7 Pairs of Contrasting Traits in *Pisum sativum* (14 True-Breeding Lines):

S.No	Character Studied	Dominant Trait	Recessive Trait
1	Stem Height	Tall (T)	Dwarf (t)
2	Flower Colour	Violet / Purple (W)	White (w)
3	Flower Position	Axial (A)	Terminal (a)
4	Pod Shape	Inflated / Full (I)	Constricted (i)
5	Pod Colour	Green (G)	Yellow (g)
6	Seed Shape	Round (R)	Wrinkled (r)
7	Seed / Cotyledon Colour	Yellow (Y)	Green (y)

4. Inheritance of One Gene (Monohybrid Cross) & Mendelian Laws

Mendel crossed pure-breeding tall (TT) pea plants with pure-breeding dwarf (tt) plants. The entire first filial generation (F_1) was uniformly tall (Tt). Upon selfing the F_1 plants, the unexpressed dwarf phenotype reappeared in the F_2 generation without any intermediate blending.

Phenotypic Ratio in F_2 = 3 Tall : 1 Dwarf (3 : 1)

Genotypic Ratio in F_2 = 1 TT : 2 Tt : 1 tt (1 : 2 : 1)

Mathematical / Binomial Derivation of Monohybrid Cross:

The gametic frequencies of alleles T and t produced by F_1 (Tt) are equal ($\frac{1}{2} T + \frac{1}{2} t$).

The F_2 genotypic distribution is the expansion of the binomial square:

$$(ax + by)^2 = (\frac{1}{2} T + \frac{1}{2} t)^2 = \frac{1}{4} TT + \frac{1}{2} Tt + \frac{1}{4} tt \rightarrow \text{Ratio} = 1 : 2 : 1$$

Mendel's Principles / Laws Derived from Monohybrid Cross:

1. Law of Dominance (First Law):

- Characters are discrete and controlled by particulate units called *factors* occurring in pairs.
- In a dissimilar pair of factors (heterozygous condition, e.g., Tt), one member dominates (dominant allele T) over

the other (recessive allele t), which remains unexpressed. Explains why F_1 expresses only one parental trait while F_2 expresses both in 3:1 ratio.

2. Law of Segregation / Purity of Gametes (Second Law - Universally Applicable):

- Alleles do not blend or intermix in a heterozygote. During meiosis / gamete formation, homologous chromosomes and their paired alleles segregate randomly, ensuring each haploid gamete receives only **one** of the two alleles with equal (50%) probability.

5. Test Cross vs. Back Cross & Punnett Square

Punnett Square (Checkerboard): Formulated by British geneticist **Reginald C. Punnett**. A graphical matrix utilized to determine all possible combinations of maternal and paternal gametes to compute genotypic and phenotypic probabilities of offspring.

Types of Crosses:

- Back Cross:** Cross of an F_1 hybrid with *either* of its parental genotypes (dominant or recessive).
- Out Cross:** Cross of F_1 hybrid (Tt) with homozygous dominant parent (TT) → All offspring are phenotypically dominant (100% Tall).
- Test Cross:** Cross of an individual showing a dominant phenotype (genotype unknown: homozygous TT or heterozygous Tt) with the **homozygous recessive** parent (tt).

Significance & Ratios of Test Cross:

- Case 1 (If unknown is Homozygous TT):**
 $TT \times tt \rightarrow 100\% Tt$ (All Tall). Confirms homozygous purity.
- Case 2 (If unknown is Heterozygous Tt):**
 $Tt \times tt \rightarrow 50\% Tt$ (Tall) : $50\% tt$ (Dwarf) → **1 : 1**. Confirms hybrid nature.
- Dihybrid Test Cross Ratio:** $RrYy \times rryy \rightarrow 1 : 1 : 1 : 1$

6. Non-Mendelian Inheritance & Gene Interactions

A. Incomplete Dominance (Partial / Intermediate Inheritance)

Discovered by **Carl Correns (1903)** in *Mirabilis jalapa* (Four O'clock plant) and *Antirrhinum majus* (Snapdragon / Dog flower). Also seen in Andalusian fowls (Black × White → Blue fowls).

- The dominant allele fails to exert complete dominance over the recessive allele, yielding an intermediate phenotype in heterozygotes (Rr = Pink flowers).

Cross: Red (RR) × White (rr)	F_1 Generation: All Pink (Rr)	F_2 Phenotypic Ratio	F_2 Genotypic Ratio
Selfing: $Rr \times Rr$	1 Red (RR) : 2 Pink (Rr) : 1 White (rr)	1 : 2 : 1	1 : 2 : 1

Key NEET Point: In Incomplete Dominance, the F_2 **Phenotypic ratio equals Genotypic ratio (1:2:1)**, deviating from Mendel's 3:1 phenotypic ratio.

B. Co-Dominance & Multiple Allelism

Co-Dominance: A phenomenon where both alleles in a heterozygote express themselves equally and simultaneously without blending (e.g., AB blood group, Roan coat color in cattle, Sickle cell carrier $Hb^A Hb^S$).

Multiple Allelism: Occurrence of more than two allelic variants of a gene occupying the same locus in a population (can only be studied in population genetics).

ABO Blood Group in Humans (Discovered by Karl Landsteiner; Bernstein elucidated inheritance):

- Controlled by gene *I* (Isoagglutininogen) having 3 alleles: I^A , I^B , and i .
- I^A and I^B produce distinct glycosyltransferase enzymes adding specific sugar polymers (A and B antigens) on RBC membrane surface. i produces no antigen.
- I^A and I^B are completely dominant over i , but I^A and I^B are **co-dominant** with each other (Genotype $I^A I^B \rightarrow$ Blood group AB).
- Total Genotypes = 6; Total Phenotypes = 4 [Formula: $n(n+1)/2$, where $n=3 \rightarrow 3(4)/2 = 6$].

Genotype	Allele from P1	Allele from P2	Antigen on RBC	Antibody in Plasma	Blood Group
$I^A I^A / I^A i$	I^A	I^A or i	Antigen A	Anti-B	A
$I^B I^B / I^B i$	I^B	I^B or i	Antigen B	Anti-A	B
$I^A I^B$	I^A	I^B	Antigen A & B (Co-dominance)	None	AB (Universal Recipient)
ii	i	i	Neither A nor B	Anti-A & Anti-B	O (Universal Donor)

C. Comprehensive Comparison Table: Dominance vs Incomplete Dominance vs Co-Dominance

Feature	Complete Dominance	Incomplete Dominance	Co-Dominance
F₁ Phenotype	Resembles dominant parent	Intermediate between parents	Resembles both parents simultaneously
Allelic Interaction	Dominant completely masks recessive	Dominant allele is partially functional	Both alleles produce full products independently
F₂ Phenotypic Ratio	3 : 1	1 : 2 : 1	1 : 2 : 1
F₂ Genotypic Ratio	1 : 2 : 1	1 : 2 : 1	1 : 2 : 1
Classical Examples	Pea height ($TT \times tt$)	Snapdragon flower color (Rr)	Human AB blood group, Roan cattle (RW)

D. Pleiotropy vs Polygenic Inheritance

Pleiotropy (One Gene → Multiple Traits):

A single gene influences multiple unrelated phenotypic traits by controlling a central biochemical/metabolic pathway.

- **Phenylketonuria (PKU):** Mutation on chromosome 12 → deficiency of *phenylalanine hydroxylase* → accumulation of phenylpyruvate → mental retardation, hypopigmentation of skin and hair.
- **Starch synthesis in Pea:** Gene *B* controls seed starch branching and seed shape. *BB* (round, large starch grains), *bb* (wrinkled, small grains), *Bb* (round seeds, but intermediate size starch grains → shows pleiotropy + incomplete dominance).
- **Sickle Cell Gene:** Affects RBC shape, hemoglobin solubility, and confers malaria resistance.

Polygenic Inheritance (Many Genes → One Trait):

A quantitative trait controlled by three or more pairs of polygenes showing continuous variation. The phenotype reflects the additive/cumulative effect of dominant alleles and is influenced by the environment.

- **Human Skin Colour (Davenport / Galton):**

Governed by 3 genes (*A, B, C*).

- *AABBCC*: Very Dark / Black (Maximum melanin)
 - *aabbcc*: Very Light / White (Minimum melanin)
 - *AaBbCc*: Intermediate / Mulatto
- Phenotypic ratio in F_2 follows a bell-shaped curve: **1 : 6 : 15 : 20 : 15 : 6 : 1** (64 combinations).

- **Kernel colour in wheat:** First experimentally proven by **Nilsson-Ehle** (ratio 1:4:6:4:1).

7. Inheritance of Two Genes (Dihybrid Cross) & Chromosomal Theory

Mendel crossed homozygous round yellow seeded plants (***RRYY***) with wrinkled green seeded plants (***rryy***). The F_1 was all ***RrYy*** (Round Yellow). Selfing of F_1 produced 16 combinations in F_2 .

F_2 Phenotypic Ratio = 9 Round Yellow : 3 Round Green : 3 Wrinkled Yellow : 1 Wrinkled Green (9 : 3 : 3 : 1)

F_2 Genotypic Ratio = 1:2:1:2:4:2:1:2:1 (9 distinct genotypes)

Law of Independent Assortment (Third Law):

When two pairs of traits are combined in a hybrid, segregation of one pair of characters is completely independent of the segregation of the other pair of characters during gametogenesis.

Mathematical Product Rule: (3 Round : 1 Wrinkled) × (3 Yellow : 1 Green) = **9 Round-Yellow : 3 Wrinkled-Yellow : 3 Round-Green : 1 Wrinkled-Green.**

Rediscovery of Mendelism (1900):

Mendel's work remained unrecognized from 1865 to 1900 due to poor scientific communication, unconventional use of mathematics in biology, and lack of cytological proof for discrete "factors". In 1900, three botanists independently rediscovered his principles: **1. Hugo de Vries (Holland), 2. Carl Correns (Germany), 3. Erich von Tschermak (Austria).**

Chromosomal Theory of Inheritance (Sutton and Boveri, 1902):

Walter Sutton and **Theodor Boveri** noticed complete parallelism between the behavior of chromosomes and the behavior of Mendelian genes during meiosis and fertilization:

Behavior of Mendelian Genes (Factors)	Behavior of Chromosomes
Occur in pairs in diploid somatic cells.	Occur in homologous pairs in diploid somatic cells.
Segregate during gamete formation so only 1 allele enters a gamete.	Homologous pairs segregate during Anaphase I of Meiosis.
Independent pairs assort independently of one another.	Non-homologous chromosome bivalents align and assort independently at Metaphase I.

8. Linkage and Genetic Recombination (T.H. Morgan)

Thomas Hunt Morgan (the *Father of Experimental Genetics*) established the experimental validation of chromosomal theory using the fruit fly, *Drosophila melanogaster*.

Why Morgan Chose *Drosophila*:

- Reared easily on simple synthetic medium in lab bottles.
- Extremely short life cycle (~2 weeks / 14 days).
- A single mating produces hundreds of progeny flies.
- Distinct sexual dimorphism (females larger, males smaller with dark ovipositor/abdomen).
- Possesses only 4 pairs of chromosomes with high distinct morphological variations visible under low-power stereomicroscope.

Key Findings on Linkage & Crossing Over:

- **Linkage:** The physical association of two or more genes located on the same chromosome. Linked genes do not assort independently and deviate from the 9:3:3:1 ratio.
- **Recombination:** The generation of non-parental gene combinations resulting from physical crossing over between homologous non-sister chromatids during pachytene stage of Meiosis I.
- **Distance vs Linkage:** Strength of linkage is inversely proportional to distance between genes. Crossing over frequency is directly proportional to map distance.

Morgan's Dihybrid Cross Results in *Drosophila*:

- **Cross A (Body colour & Eye colour - *y* and *w*):** Closely linked on X chromosome → **98.7% Parental types** and only **1.3% Recombinants** (Tightly linked).
- **Cross B (Eye colour & Wing size - *w* and *m*):** Loosely linked on X chromosome → **62.8% Parental types** and **37.2% Recombinants**.
- **Alfred Sturtevant (Morgan's Student):** Utilized recombination frequencies (RF) to construct the first **Genetic Maps / Chromosome Maps**. $1\% \text{ Recombination frequency} = 1 \text{ CentiMorgan (cM)} = 1 \text{ Map Unit (m.u.)}$. Extensively utilized today in Human Genome Project.

9. Mechanisms of Sex Determination

The chromosomal basis of sex was initiated by **Hermann Henking (1891)**, who traced a specific nuclear structure through spermatogenesis in insects, designating it the '**X-body**'. Later recognized as the X chromosome.

Mechanism Type	Male Genotype	Female Genotype	Organism Examples
XX - XY Type (Male Heterogamety)	XY (Heterogametic: 50% X, 50% Y sperms)	XX (Homogametic: 100% X ova)	Humans, <i>Drosophila</i> , most mammals
XX - XO Type (Male Heterogamety)	XO (50% X sperms, 50% null sperms)	XX (Homogametic: all ova carry X)	Grasshoppers, Cockroaches, Bugs
ZZ - ZW Type (Female Heterogamety)	ZZ (Homogametic: all sperms carry Z)	ZW (Heterogametic: 50% Z, 50% W ova)	Birds, Fowls, Reptiles, certain Fishes
ZZ - ZO Type (Female Heterogamety)	ZZ (Homogametic)	ZO (Heterogametic)	Butterflies, Moths

Haplodiploid Sex Determination in Honey Bees (*Apis mellifera*):

- Sex is determined by the number of chromosome sets received by an individual.
- **Females (Queen / Worker):** Formed by sexual syngamy of sperm ($n=16$) and egg ($n=16$) → **Diploid ($2n = 32$ chromosomes)**.
- **Males (Drones):** Formed unfertilized from haploid eggs ($n=16$) via **Parthenogenesis** → **Haploid ($n = 16$ chromosomes)**.
- *Unique Fact:* Drones produce sperms by **mitosis** (not meiosis). Consequently, male honeybees **have no father and cannot have sons, but they have a grandfather and can have grandsons!**

10. Mutation & Pedigree Analysis

Mutation: Sudden, heritable alteration in DNA nucleotide sequences or chromosomal structure/number leading to changes in genotype and phenotype. Agents inducing mutations are **Mutagens** (Physical: UV, X-rays, Gamma rays; Chemical: Nitrous acid, EMS, Alkylating agents; Biological: Oncoviruses, transposons).

- **Point Mutation:** Alteration in a single base pair of DNA (e.g., Sickle-cell anemia: GAG → GUG transition).
- **Frameshift Mutation:** Deletion or insertion of one or more base pairs that shifts the entire reading frame of mRNA triplets (e.g., Thalassemia).

Pedigree Analysis (Genetic Family Tree):

Systematic study of the inheritance of specific genetic traits, abnormalities, or diseases across multiple generations of human families using standardized symbols.

Standard Pedigree Symbols:

-  Square = Male | **&Circle;** Circle = Female |  Diamond = Sex unspecified
- Solid/Shaded Symbol = Affected individual | Open Symbol = Normal individual
- Horizontal Line between square and circle = Mating / Marriage | Double Horizontal Line = Consanguineous mating (inbreeding)
- Half-shaded / Dot in center = Heterozygous carrier | Line through symbol = Deceased

11. Human Genetic Disorders (Comprehensive Classification)

Human genetic disorders are classified into two broad categories: **Mendelian Disorders** and **Chromosomal Disorders**.

A. Mendelian Disorders (Gene Mutations)

Disorder Name	Inheritance Pattern	Gene / Chromosome	Etiology, Pathology & Clinical Manifestations
Sickle Cell Anaemia	Autosomal Recessive ($Hb^A Hb^S$)	Chromosome 11 (<i>HBB</i> gene)	Point mutation substituting Glutamic acid (GAG) with Valine (GUG) at the 6th position of β -globin chain. Under low oxygen tension, HbS polymerizes into sickle-shaped RBCs causing hemolytic anemia, microvascular occlusion, and organ infarction. Heterozygotes show malaria resistance.
Thalassemia	Autosomal Recessive (Quantitative defect)	α : Chromosome 16 (<i>HBA1</i> , <i>HBA2</i>) β : Chromosome 11 (<i>HBB</i>)	Imbalance in globin chain synthesis due to gene deletion/mutation. α-Thalassemia has deficient α -chain; β-Thalassemia (Cooley's anemia) has deficient β -chain. Results in excess unpaired chains, microcytic hypochromic anemia, bone deformities, and hepatosplenomegaly.
Phenylketonuria (PKU)	Autosomal Recessive (Inborn error of metabolism)	Chromosome 12	Deficiency of hepatic <i>phenylalanine hydroxylase</i> . Phenylalanine cannot convert to Tyrosine and converts to phenylpyruvic acid. Causes irreversible mental retardation, microcephaly, and hypopigmentation.
Haemophilia (Royal Disease)	X-Linked Recessive (Criss-cross inheritance)	X Chromosome	Deficiency of blood clotting Factor VIII (Haemophilia A) or Factor IX (Haemophilia B / Christmas disease). Simple cuts lead to non-stop fatal hemorrhages. Transmitted from carrier mother ($X^H X^h$) to 50% of sons. Queen Victoria was a prominent pedigree carrier.
Colour Blindness	X-Linked Recessive (Daltonism)	X Chromosome	Mutation in red/green cone photopigments. Inability to distinguish red and green colours. Affects ~8% human males and only ~0.4% females because female requires two defective X alleles ($X^c X^c$).
Myotonic Dystrophy	Autosomal Dominant	Chromosome 19 (<i>DMPK</i> gene)	Trinucleotide repeat expansion (CTG). Progressive muscle wasting, myotonia (inability to relax muscles after contraction), cataracts, cardiac defects. Vertical transmission across every generation.
Cystic Fibrosis	Autosomal Recessive	Chromosome 7 (<i>CFTR</i> gene)	Defective chloride channel glycoprotein $\rightarrow$ thick, viscous mucus clogging pancreatic ducts, respiratory passages, leading to chronic infections and malabsorption.

Disorder Name	Inheritance Pattern	Gene / Chromosome	Etiology, Pathology & Clinical Manifestations
Hypertrichosis	Y-Linked (Holandric trait)	Y Chromosome	Presence of excessive hair follicles on ear pinna. Strictly transmitted from father to 100% sons (Male-limited transmission).

B. Chromosomal Disorders (Aneuploidy & Polyploidy)

Arise due to non-disjunction (failure of chromatid/homologous chromosome separation) during meiotic cell division:

- **Aneuploidy:** Loss or gain of one or few chromosomes [$2n - 1$ (Monosomy), $2n + 1$ (Trisomy), $2n - 2$ (Nullisomy)].
- **Polyploidy:** Increase in whole sets of chromosomes ($3n$, $4n$) due to cytokinesis failure after telophase (common in plants).

Disorder Name	Karyotype & Genetic Basis	Salient Clinical Features & Diagnostic Symptoms
Down's Syndrome (Trisomy 21)	47, XX,+21 or 47, XY, +21 (Autosomal Trisomy 21) First described by Langdon Down (1866)	• Short stature with rounded flat facial profile and broad forehead. • Characteristically furrowed / protruding tongue and partially open mouth. • Broad palm with single continuous transverse crease (Simian crease) and dermatoglyphic loops. • Congenital cardiovascular defects, stunted physical, psychomotor, and mental development.
Klinefelter's Syndrome	47, XXY (Allosomal / Sex-Chromosomal Trisomy in Males)	• Overall tall, eunuchoid male phenotype with underdeveloped, small atrophic testes. • Sterile male with azoospermia (absence of sperm production). • Development of breast tissue (Gynaecomastia), sparse body hair, and feminine pitch/fat distribution.
Turner's Syndrome	45, XO (45, X) (Allosomal / Sex-Chromosomal Monosomy in Females)	• Phenotypically female, but sterile due to rudimentary/streak ovaries. • Short stature, low posterior hairline, shield-shaped thorax with widely spaced nipples. • Prominent webbing of the neck, lymphedema of extremities, absence of secondary sexual characteristics, primary amenorrhea.
Cri-du-Chat Syndrome ("Cat's Cry")	46, 5p- (Partial Monosomy / Deletion of short arm 'p' of chromosome 5)	• High-pitched mewing cry resembling a kitten due to abnormal laryngeal hypoplasia. • Microcephaly, moon-shaped round face, hypertelorism, severe intellectual disability, low birth weight.

12. High-Yield Conceptual Problems & Exam Exercises (with Solutions)

Exercise 1: A child has blood group 'O'. If father has blood group 'A' and mother has blood group 'B', deduce parental genotypes and possible phenotypes of future children.

Solution: Since child is ii (O group), each parent must donate one 'i' allele. Father's genotype is $I^A i$ (heterozygous A) and Mother's genotype is $I^B i$ (heterozygous B). Future offspring genotypes can be $I^A I^B$ (AB), $I^A i$ (A), $I^B i$ (B), ii (O) in equal $1 : 1 : 1 : 1$ ratio (25% probability each).

Exercise 2: A carrier haemophilic woman ($X^H X^h$) marries a normal man ($X^H Y$). What percentage of their male progeny will be haemophilic?

Solution: Male gametes (X^H , Y); Female gametes (X^H , X^h). Sons receive Y from father and either X^H or X^h from mother $\rightarrow$ 50% Normal sons ($X^H Y$) and **50% Haemophilic sons ($X^h Y$)**. Among all total offspring, 25% are haemophilic.

Exercise 3: In a test cross of dihybrid $AaBb \times aabb$, if recombinant percentage is 18%, what is the distance between genes A and B?

Solution: 1% Recombination Frequency = 1 CentiMorgan (cM) = 1 Map Unit (m.u.). Since RF is 18%, the genetic map distance between genes A and B is exactly **18 cM (or 18 map units)**.

BAIO LEARNING SYSTEM — SUMMARY MASTER SHEET

- Mendel's Laws: Dominance & Segregation (Monohybrid 3:1 / 1:2:1), Independent Assortment (Dihybrid 9:3:3:1).
- Exceptions: Incomplete Dominance (1:2:1), Co-dominance (ABO blood group), Linkage (Morgan in *Drosophila*).
- Sex Determination: XX-XY (Humans), XX-XO (Insects), ZZ-ZW (Birds), Haplodiploid (Honeybee - 32/16).
- Point Mutation: Sickle Cell (GAG $\rightarrow$ GUG, Glu $\rightarrow$ Val at pos 6 of β -chain).
- Aneuploidies: Down's (21 Trisomy), Klinefelter's (47, XXY), Turner's (45, XO), Cri-du-chat (5p deletion).